

Solution 3: VaR Calculations

ME317: Statistical Methods for Risk Management

Question 1 — Linearity (translation & scaling) of VaR

Show that $\text{VaR}_\alpha(aL + b) = a \text{VaR}_\alpha(L) + b$ for $a > 0$, $b \in \mathbb{R}$.

Recall, for a loss $L \sim F_L$,

$$\text{VaR}_\alpha(L) := \inf\{l \in \mathbb{R} : F_L(l) \geq \alpha\} = \inf\{l \in \mathbb{R} : \mathbb{P}(L > l) \leq 1 - \alpha\},$$

where α is close to 1 (e.g. $\alpha = 0.95, 0.99$).

Proof.

$$\begin{aligned} \text{VaR}_\alpha(aL + b) &= \inf\{l \in \mathbb{R} : \mathbb{P}(aL + b > l) \leq 1 - \alpha\} \\ &= \inf\left\{l \in \mathbb{R} : \mathbb{P}\left(L > \frac{l-b}{a}\right) \leq 1 - \alpha\right\} && (a > 0) \\ &= \inf\{al' + b : \mathbb{P}(L > l') \leq 1 - \alpha\} && \left(l' = \frac{l-b}{a}, l = al' + b\right) \\ &= a \inf\{l' \in \mathbb{R} : \mathbb{P}(L > l') \leq 1 - \alpha\} + b \\ &= a \text{VaR}_\alpha(L) + b. \quad \blacksquare \end{aligned}$$

Question 2 — Sub-additivity of VaR (diversification effect)

In general $\text{VaR}_\alpha(L_1 + L_2) \leq \text{VaR}_\alpha(L_1) + \text{VaR}_\alpha(L_2)$ does *not* hold, but it does for independent normal risks.

Suppose $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ with $X \perp Y$. Then

$$\text{VaR}_\alpha(X + Y) \leq \text{VaR}_\alpha(X) + \text{VaR}_\alpha(Y).$$

Proof. Since $X \perp Y$, $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Writing a normal as $\mu + \sigma Z$ with $Z \sim N(0, 1)$ and using Question 1,

$$\begin{aligned} \text{VaR}_\alpha(X + Y) &= (\mu_1 + \mu_2) + \sqrt{\sigma_1^2 + \sigma_2^2} \text{VaR}_\alpha(Z) \\ &\leq (\mu_1 + \mu_2) + (\sigma_1 + \sigma_2) \text{VaR}_\alpha(Z) && (\sqrt{\sigma_1^2 + \sigma_2^2} \leq \sigma_1 + \sigma_2, \text{VaR}_\alpha(Z) \geq 0) \\ &= (\mu_1 + \sigma_1 \text{VaR}_\alpha(Z)) + (\mu_2 + \sigma_2 \text{VaR}_\alpha(Z)) \\ &= \text{VaR}_\alpha(X) + \text{VaR}_\alpha(Y). \quad \blacksquare \end{aligned}$$

(The step uses $\text{VaR}_\alpha(Z) = \Phi^{-1}(\alpha) \geq 0$, which holds for $\alpha \geq \frac{1}{2}$.)

The risk of holding two positions together should not be worse than simply adding up their risks individually.

Question 3 — One-day VaR of a single position

$\lambda = 5$ shares, $S_0 = 100$, one-day log-return $X_1 = \log(S_1/S_0) \sim N(0, 0.1^2)$, loss $L = -\lambda(S_1 - S_0)$.

(1) **The loss as a function of X_1 .** With log-return $X_1 := \log(S_1/S_0)$, so that $S_1/S_0 = e^{X_1}$,

$$L = -\lambda(S_1 - S_0) = -\lambda S_0 \left(\frac{S_1}{S_0} - 1 \right) = -\lambda S_0 (e^{X_1} - 1) = -500(e^{X_1} - 1) =: f(X_1).$$

(2) **Range of L .** $e^{X_1} \in (0, \infty) \Rightarrow e^{X_1} - 1 \in (-1, \infty) \Rightarrow L \in (-\infty, 500)$. (The long position cannot lose more than its value $\lambda S_0 = 500$.)

(3) **Compute $\text{VaR}_{0.99}(L)$, using $F_X^{-1}(0.99) \approx 0.23$.** $\text{VaR}_\alpha(L)$ satisfies $\mathbb{P}(L \leq \text{VaR}_\alpha(L)) = \alpha$. Let $q := \text{VaR}_\alpha(L)$. Then

$$\begin{aligned} \mathbb{P}(-500(e^{X_1} - 1) \leq q) &= \alpha \iff \mathbb{P}\left(e^{X_1} \geq 1 - \frac{q}{500}\right) = \alpha \\ &\iff \mathbb{P}\left(X_1 \geq \log\left(1 - \frac{q}{500}\right)\right) = \alpha. \end{aligned}$$

Hence, by symmetry of the normal, $\log\left(1 - \frac{q}{500}\right) = F_X^{-1}(1 - \alpha) = F_X^{-1}(0.01) = -0.23$, so

$$q = (1 - e^{-0.23}) \times 500 \approx 102.733.$$

There is a 1% chance the loss exceeds 102.733.

Question 4 — 100-day loss: exact vs. linearised

Hold the same portfolio for $T = 100$ trading days; daily log-returns $X_t \stackrel{\text{iid}}{\sim} N(0, 0.1^2)$. Loss L_{100} .

(1) **Exact $\text{VaR}_{0.99}(L_{100})$.** The same form carries over directly: the cumulative return factors into a product of daily ratios,

$$L_T = -\lambda(S_T - S_0) = -\lambda S_0 \left(\frac{S_T}{S_0} - 1 \right) = -\lambda S_0 \left(\frac{S_T}{S_{T-1}} \cdots \frac{S_1}{S_0} - 1 \right) = -\lambda S_0 \left(e^{\sum_{t=1}^T X_t} - 1 \right),$$

using $\frac{S_T}{S_0} = \prod_{t=1}^T \frac{S_t}{S_{t-1}} = \prod_{t=1}^T e^{X_t} = e^{\sum_{t=1}^T X_t}$. For $T = 100$,

$$L_{100} = -500 \left(e^{\sum_{t=1}^{100} X_t} - 1 \right), \quad \sum_{t=1}^{100} X_t \sim N(0, 100 \cdot 0.1^2) = N(0, 1).$$

Writing $Z := \sum_{t=1}^{100} X_t \sim N(0, 1)$, so $L_{100} = -500(e^Z - 1)$. With $q := \text{VaR}_{0.99}(L_{100})$,

$$\mathbb{P}(-500(e^Z - 1) \leq q) = 0.99 \iff \mathbb{P}\left(Z \geq \log\left(1 - \frac{q}{500}\right)\right) = 0.99,$$

so $\log\left(1 - \frac{q}{500}\right) = -\Phi^{-1}(0.99)$ and

$$q = 500 \left(1 - e^{-\Phi^{-1}(0.99)} \right) \approx 500(1 - e^{-2.326}) \approx 451.2.$$

(2) **Linearised** $\text{VaR}_{0.99}(L_{100}^\Delta)$. Using $e^x \approx 1 + x \Rightarrow e^x - 1 \approx x$, the linearised loss is $L_T^\Delta(x) = -\lambda S_0 x = -500x$, so

$$L_{100}^\Delta = -500 \sum_{t=1}^{100} X_t \stackrel{d}{=} -500 Z, \quad Z \sim N(0, 1), \quad \text{i.e.} \quad L_{100}^\Delta \sim N(0, 500^2).$$

By the linearity of VaR (Question 1),

$$\text{VaR}_\alpha(L_{100}^\Delta) = 500 \text{VaR}_\alpha(Z) = 500 \Phi^{-1}(\alpha), \quad \text{VaR}_{0.99}(L_{100}^\Delta) = 500 \Phi^{-1}(0.99) \approx 1163.$$

Remark. The linearised VaR (≈ 1163) far exceeds the exact VaR (≈ 451), and even exceeds the maximum possible loss of 500. Over a long horizon (here the 100-day return has standard deviation 1), the approximation $e^x - 1 \approx x$ breaks down: it ignores that a long position's loss is bounded, and so badly overstates the risk.